AI-BASED REAL TIME EMOTION DETECTION AND PERSONALIZED MUSIC AND ACTIVITY RECOMMENDATION SYSTEM

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

The work embodied in the present project report entitled "**AI-BASED REAL TIME EMOTION DETECTION AND PERSONALIZED MUSIC AND ACTIVITY RECOMMENDATION SYSTEM**" has been carried out by the students **DHARSHINI S S, DHEETSHIKA R, INDHUMATHI S.** The work reported herein is original and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

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ABSTRACT

This project proposes an intelligent emotion detection system that analyzes facial expressions to identify user emotions such as happiness, sadness, and anger. The system uses computer vision and machine learning techniques to detect emotions through facial recognition. Based on the detected emotion, it recommends suitable activities such as music playlists, gaming activities, and book reading suggestions. The system also provides wellness feedback and personalized recommendations to improve user mood. Additionally, it analyzes user activity patterns and presents the results using graphical visualization for better understanding of emotional trends.

Keywords: Emotion Detection, Facial Expression Recognition, Activity Recommendation, Machine Learning, User Behavior Analysis.

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SIGNATURE

TABLE OF CONTENTS

CHAPTER	TITLE	PAGE No.
	ABSTRACT	**iii**
	LIST OF FIGURES	**vii**
	LIST OF ABBREVIATIONS	**viii**
1	**INTRODUCTION**	**1**
	1.1 DESCRIPTION	1
	1.2 TECHNOLOGY OVERVIEW	1
	1.2.1 Artificial Intelligence	2
	1.2.2 Machine Learning	2
	1.2.3 Computer Vision	3
	1.3 PROBLEM STATEMENT	3
	1.4 OBJECTIVE	4
	1.5 IMPLICATION	5
2	**LITERATURE SURVEY**	**6**
3	**EXISTING SYSTEM**	**12**
4	**PROBLEMS IDENTIFIED**	**14**
5	**PROPOSED SYSTEM**	**15**
6	**SYSTEM REQUIREMENTS**	**17**
	6.1 HARDWARE REQUIREMENTS	17
	6.2 SOFTWARE REQUIREMENTS	18
7	**SYSTEM IMPLEMENTATIONS**	**21**
	7.1 LIST OF MODULES	21
	7.2 MODULES DESCRIPTION	21
	7.2.1 User Input Module	21
	7.2.2 AI Emotion Detection Module	22

	7.2.3 Emotion classification Module	23
	7.2.4 Smart Recommendation Engine Module	23
	7.2.5 Activity Recommendation Module	24
	7.2.6 User Data and Analytics Module	25
8	**SYSTEM TESTING**	**27**
	8.1 USER INTERFACE TESTING	27
	8.2 INTEGRATION TESTING	27
	8.3 SYSTEM TESTING	28
	8.4 PERFORMANCE TESTING	28
	8.5 ACCURACY TESTING	29
	8.6 FUNCTIONAL TESTING	29
9	**RESULTS AND DISCUSSION**	**31**
10	**CONCLUSION AND FUTURE WORK**	**33**
	10.1 CONCLUSION	33
	10.2 FUTURE ENHANCEMENT	34
	APPENDIX A - SOURCE CODE	**35**
	APPENDIX B - SCREENSHOTS	**43**
	REFERENCES	**46**

LIST OF FIGURES

FIGURE NO.	TITLE	PAGE NO.
3.1	Existing System	13
5.1	Proposed Diagram	16
B.1	Start Detection	43
B.2	Text Input	43
B.3	Emotion Analysis	44
B.4	Music Recommendation	44
B.5	Feedback	45

LIST OF ABBREVIATIONS

CNN	- Convolutional Neural Network
ML	- Machine Learning
UAT	- User Acceptance Testing
GPU	- Graphics Processing Unit
CPU	- Control Processing Unit
AI	- Artificial Intelligence
API	- Application Programming Interface
RAM	- Random Access Memory
IDE	- Integrated Development Environment
FER	- Facial Emotion Recognition
SRE	- Smart Recommendation Engine

CHAPTER 1

INTRODUCTION

1.1 DESCRIPTION

The emotion detection and activity recommendation system is designed to analyze user emotions through facial expressions and provide suitable activities accordingly. The system captures the user's facial image through a camera and processes it using computer vision techniques. A trained machine learning model analyzes facial features such as eyes, mouth, and facial structure to determine the user's emotion. The system can identify emotions such as happiness, sadness, anger, and neutral mood. After detecting the emotion, the system suggests suitable activities that may help the user maintain or improve their mood. These activities include music playlist recommendations, gaming activities, book reading suggestions, and wellness tips. The system also records user activity choices and stores them for future analysis. By comparing current activities with previous activities, the system can identify user behavior patterns. This information helps the system generate better recommendations in the future. The analysis results are displayed in graphical form for easy understanding.

1.2 TECHNOLOGY OVERVIEW

The proposed system is built using advanced technologies such as Artificial Intelligence, Machine Learning, and Computer Vision. It utilizes a Convolutional Neural Network (CNN) model to detect and classify human emotions from facial expressions. Computer Vision techniques are used to capture and process real-time images through a webcam. Machine Learning algorithms help improve the accuracy of emotion recognition by learning from

data. The system also includes a smart recommendation engine that suggests activities based on the detected emotional state.

1.2.1 Artificial Intelligence

Artificial Intelligence (AI) plays an important role in improving the intelligence and security of modern software systems. AI enables computers to simulate human intelligence and perform tasks such as learning, analyzing patterns, and making decisions. In this project, AI techniques are used to improve the overall security of the messaging system. The system uses AI concepts to analyze user behavior and communication patterns.

AI algorithms help the system make intelligent decisions based on the detected emotional state. The system processes facial data and interprets patterns to classify emotions accurately. AI also helps improve system performance by learning from previous data. The integration of AI allows the system to provide personalized activity recommendations. AI techniques ensure that the system adapts to user behavior and improves over time. This makes the system more efficient and reliable. Artificial Intelligence forms the foundation of the entire project.

1.2.2 Machine Learning

Machine Learning (ML) is a subset of Artificial Intelligence that allows systems to learn from data and improve their performance without explicit programming. Machine learning algorithms analyze large amounts of data and identify patterns that can be used for prediction or decision-making.

In this project, Machine Learning is used to monitor communication activities and detect suspicious behavior within the messaging system. The

ML model analyses factors such as message frequency, login behavior, and communication patterns between users. If unusual activities are detected, the system can generate alerts or restrict suspicious actions. Machine learning helps the system become more intelligent and adaptive.

1.2.3 Computer vision

Computer Vision is a technology that enables computers to interpret and understand visual information from images or videos. In this project, computer vision techniques are used to detect the user's face from the camera input. computer vision techniques are used to detect the user's face from the camera input. The system captures real-time images and identifies facial features such as eyes, nose, and mouth. These features are analyzed to determine the user's emotional state.

Computer vision algorithms help in detecting faces accurately even in different lighting conditions. Libraries such as OpenCV are commonly used for face detection and image processing. Computer vision also helps in preprocessing images before they are analyzed by the machine learning model. This improves the accuracy of emotion detection. Computer vision plays a crucial role in enabling the system to analyze facial expressions effectively.

1.3 PROBLEM STATEMENT

In today’s fast-paced world, many individuals experience emotional stress, mood fluctuations, and mental fatigue due to various personal and professional factors. However, most existing systems are not capable of understanding a user’s emotional state in real time and providing appropriate support. Traditional applications mainly rely on manual input or generic recommendations, which do not effectively address individual emotional needs.

Although emotion detection systems exist, they are mostly limited to identifying emotions and do not provide meaningful or personalized activity suggestions. There is also a lack of integration between emotion recognition and activity recommendation systems. As a result, users do not receive proper guidance to improve or manage their emotional well-being.

Therefore, there is a need to develop an intelligent system that can automatically detect user emotions using facial expressions and provide suitable activity recommendations such as music, games, reading, or relaxation techniques. The system should also analyze user behavior and adapt over time to deliver more personalized and effective suggestions.

1.4 OBJECTIVE

The main objective of this project is to develop an intelligent system that can detect human emotions using facial expression recognition techniques. The system aims to utilize Artificial Intelligence and Machine Learning algorithms to accurately identify and classify emotions such as happiness, sadness, anger, and neutral mood. Another important objective is to design a smart recommendation engine that provides personalized activity suggestions based on the detected emotional state. These activities may include music, games, book reading, and relaxation techniques to improve the user's mood and overall well-being.

The system also focuses on analyzing user behavior and activity patterns to enhance future recommendations. It aims to provide a user-friendly interface for easy interaction and ensure real-time processing of user input for quick response. Additionally, the system stores and visualizes user data to track emotional trends over time. Overall, the objective is to build an efficient and reliable system that integrates emotion detection with activity recommendation to support users in managing their emotional health.

1.5 IMPLICATION

The proposed Emotion-Based Activity Recommendation System has significant implications in improving user well-being and enhancing human-computer interaction. By detecting user emotions in real time, the system can provide immediate and personalized support through suitable activity recommendations. This helps users manage stress, improve their mood, and maintain better mental health. The system can be useful in various domains such as healthcare, education, and entertainment, where understanding user emotions is important.

In the healthcare sector, it can assist in monitoring emotional conditions and providing basic mental wellness support. In education, it can help students manage stress and improve focus by suggesting appropriate activities. In entertainment, it enhances user experience by offering personalized content based on emotions. The system also promotes the use of Artificial Intelligence and Machine Learning in real-world applications. Overall, this project contributes to the development of intelligent systems that can understand human emotions and respond effectively, making technology more adaptive and user-centric.

CHAPTER 2

LITERATURE SURVEY

2.1 SMART EMOTION BASED ACTIVITY RECOMMENDATION SYSTEM

This research paper presents a smart emotion-based activity recommendation system that identifies user emotions and provides suitable activities to improve the user's mental state. The system uses facial expression recognition techniques to detect emotions such as happiness, sadness, and anger. Based on the detected emotion, the system suggests appropriate activities like listening to music, watching videos, or engaging in relaxing tasks. Machine learning algorithms are used to analyze facial features and classify emotions accurately. The proposed system aims to improve user engagement and emotional well-being. The study highlights the importance of integrating emotion recognition with recommendation systems. The system also focuses on providing personalized suggestions for users. This research demonstrates how intelligent systems can support users by understanding their emotional conditions. It provides a foundation for developing more advanced emotion-aware applications.

2.2 MUSIC RECOMMENDATION SYSTEM BASED ON FACIAL EMOTION RECOGNITION

This study focuses on developing a music recommendation system that uses facial emotion recognition to suggest appropriate music to users. The system captures facial expressions through a webcam and analyzes them using computer vision techniques. Machine learning models are trained to classify emotions such as happy, sad, angry, and neutral. Based on the detected emotion, the system recommends suitable songs from a music database. The

purpose of this system is to enhance the user's listening experience by providing personalized music recommendations. The research emphasizes the role of emotion recognition in improving entertainment applications. It also demonstrates how music can influence and regulate human emotions. The study shows that integrating facial recognition technology with music recommendation systems can improve user satisfaction. This approach makes music selection more dynamic and personalized.

2.3 EMOTION BASED MUSIC RECOMMENDATION SYSTEM

This research work presents an emotion-based music recommendation system that identifies user emotions and suggests music accordingly. The system uses facial expression analysis to detect emotional states. Image processing techniques are used to extract facial features from captured images. Machine learning algorithms are then applied to classify the detected emotions. Once the emotion is identified, the system recommends music tracks that match the user’s mood. The system aims to improve emotional well-being by providing suitable music suggestions. The research highlights the importance of emotional intelligence in recommendation systems. It also explains how music can influence emotional balance and relaxation. The proposed system helps users discover music that matches their current emotional state. This research contributes to the development of intelligent entertainment systems.

2.4 INTELLIGENT MUSIC RECOMMENDATION SYSTEM BASED ON FACIAL EMOTION RECOGNITION

This paper introduces an intelligent music recommendation system that uses facial emotion recognition to suggest music playlists. The system detects facial expressions through a camera and processes the images using computer vision techniques. Deep learning models are used to classify emotions with

high accuracy. Based on the detected emotion, the system automatically selects appropriate songs from a predefined playlist. The goal of the system is to enhance user experience by delivering emotionally relevant music. The research highlights the benefits of integrating artificial intelligence with recommendation systems. It also demonstrates how intelligent systems can adapt to user emotions. The proposed system can be applied in entertainment platforms, smart devices, and mobile applications. The research shows that emotion-aware music systems can significantly improve user engagement.

2.5 AI-BASED EMOTION RECOGNITION AND PERSONALIZED MUSIC THERAPY

This research focuses on using artificial intelligence for emotion recognition and personalized music therapy. The system detects human emotions through facial expression analysis using machine learning techniques. Once the emotion is identified, the system recommends music that can help regulate the user's emotional state. The research highlights the importance of AI in healthcare and mental wellness applications. Music therapy is used as a method to reduce stress and improve emotional balance. The proposed system combines emotion detection with personalized therapy recommendations. The study emphasizes the potential of AI-based systems in supporting mental health. It also discusses how intelligent music therapy systems can help users manage their emotions effectively. This research provides valuable insights for developing advanced emotion-aware healthcare applications.

2.6 FACIAL EMOTION RECOGNITION USING DEEP LEARNING

This research focuses on facial emotion recognition using deep learning techniques. The system uses convolutional neural networks to analyze facial expressions and classify emotions accurately. The dataset includes various facial images representing emotions such as happiness,sadness, anger, and

surprise. The deep learning model learns facial patterns and improves classification accuracy. The research highlights the importance of automated emotion detection in intelligent systems. Facial emotion recognition can be applied in healthcare, education, and entertainment applications. The study also discusses the challenges in detecting emotions under different lighting conditions. The proposed approach demonstrates higher accuracy compared to traditional machine learning methods. This research provides a strong foundation for building intelligent emotion detection systems.

2.7 EMOTION RECOGNITION SYSTEM USING COMPUTER VISION TECHNIQUES

This research presents an emotion-aware music recommendation system that uses facial expression recognition. The system captures the user's facial image through a camera and detects emotional states using machine learning algorithms. Based on the identified emotion, the system recommends appropriate music playlists. The aim is to improve user satisfaction by providing emotionally relevant music suggestion.The system integrates emotion detection with recommendation algorithms.This approach allows user to discover music that matches users mood. The research also discusses the importance of personalized recommendation system. The proposed system improves music listening experience by adapting to emotional changes.The study demonstrates how emotion recognition can enhance entertainment applications.

2.8 EMOTION RECOGNITION SYSTEM USING COMPUTER VISION TECHNIQUES

This study focuses on developing an emotion recognition system using computer vision techniques. The system detects facial expressions from images and analyzes facial features such as eyes, eyebrows, and mouth. Image processing techniques are used to extract facial features for emotion

classification. Machine learning algorithms are used to identify emotional states. The research highlights the importance of facial expression analysis in understanding human emotions. Emotion recognition systems can be used in security, healthcare, and human-computer interaction applications. The system improves communication between humans and intelligent systems. The study also discusses methods to increase emotion detection accuracy. This research contributes to the development of intelligent emotion detection systems.

2.9 A PERSONALIZED RECOMMENDATION SYSTEM BASED ON USER EMOTIONS

This research proposes a personalized recommendation system that considers user emotions while suggesting activities or content. The system analyzes user emotional states and recommends suitable entertainment options. Machine learning algorithms are used to understand user behavior patterns. The system provides personalized suggestions based on emotional preferences. The research emphasizes the importance of emotional intelligence in recommendation systems. Personalized recommendations improve user engagement and satisfaction. The system can recommend music, movies, books, or games. It also analyzes user feedback to improve recommendations over time. The research highlights the role of emotion-aware systems in modern digital platforms. This approach improves the overall user experience.

2.10 REAL-TIME EMOTION DETECTION USING FACIAL EXPRESSION ANALYSIS

This research presents a real-time emotion detection system using facial expression analysis. The system captures live facial images using a camera and processes them using computer vision algorithms. The facial features are extracted and analyzed to detect emotional states. Machine learning techniques are used to classify emotions in real time. The research focuses on improving

the speed and accuracy of emotion recognition systems. Real-time emotion detection can be used in interactive applications and smart systems. The system can respond immediately based on user emotions. The research highlights the importance of emotion-aware technologies in human-computer interaction. This work contributes to the development of intelligent systems that respond to human emotions effectively.

CHAPTER 3

EXISTING SYSTEM

The existing emotion detection systems mainly focus on identifying human emotions using facial expression recognition techniques. These systems capture the user's facial image through a camera and analyze facial features such as eyes, eyebrows, and mouth to classify emotions. Machine learning and computer vision algorithms are commonly used to detect emotions like happiness, sadness, anger, surprise, and neutral mood. Many existing systems are designed mainly for research purposes or basic emotion recognition applications. Some systems also recommend music based on the detected emotional state. However, most of these systems provide only limited recommendations such as music playlists. They do not include multiple activity suggestions like gaming, reading books, or wellness advice. In addition, many systems do not store or analyze user activity data for future improvements. Existing systems also lack graphical visualization to analyze emotional trends and user behavior patterns. Therefore, these systems provide limited user interaction and personalization. As a result, there is a need for an improved system that not only detects emotions but also provides personalized activity recommendations and user behavior analysis.

Furthermore, existing systems often have limited accuracy when detecting emotions in different lighting conditions or facial angles. Many systems are not designed to work in real-time environments, which reduces their practical usability. These limitations highlight the need for a more efficient and intelligent emotion-based activity recommendation system.

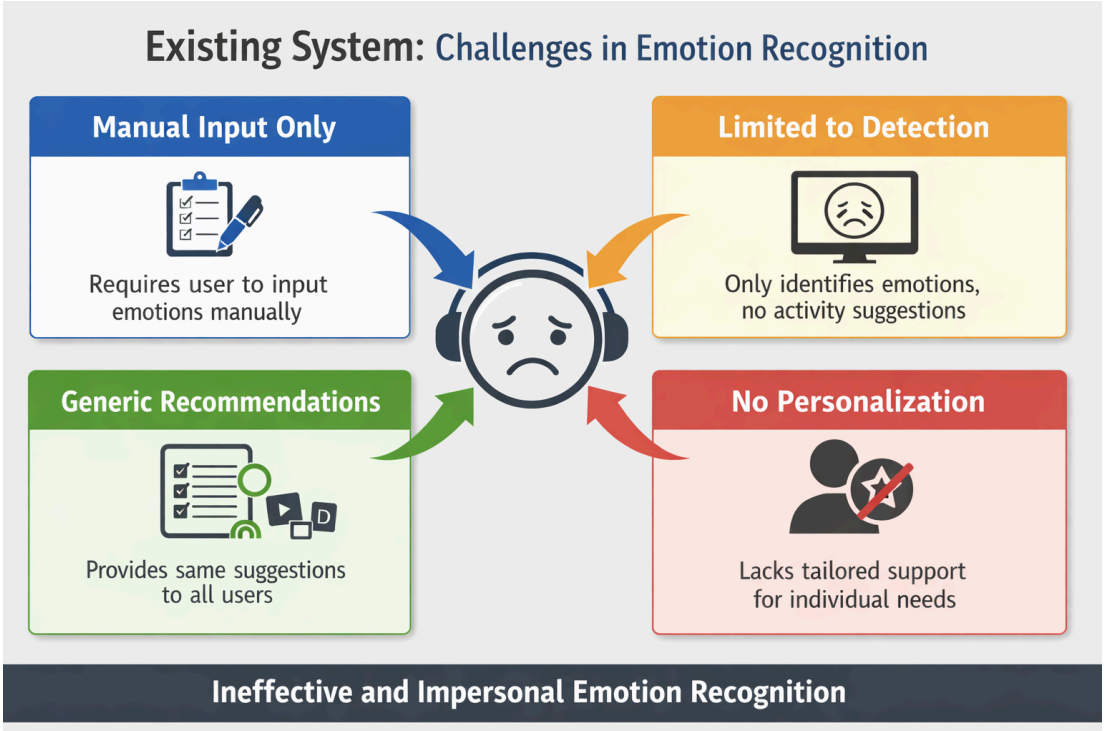

Figure 3.1 Existing System

CHAPTER 4

PROBLEMS IDENTIFIED

Existing systems mainly focus only on emotion detection and do not provide comprehensive activity recommendations for users.Most systems provide limited suggestions such as music only, without including other activities like gaming, reading books, or wellness tips.Many emotion detection systems have lower accuracy when facial images are captured in different lighting conditions or camera angles.

Existing systems often do not store user activity history, which makes it difficult to analyze user behavior over time.There is no proper analysis of user emotional patterns, so the systems cannot understand long-term emotional trends.Most systems do not provide graphical visualization to represent emotional changes and user activity analysis.Existing systems generally lack personalized recommendations, meaning the suggestions are not customized for individual users.Some emotion detection systems require large datasets and complex training models, making them difficult to implement in real-time applications.Many systems do not provide real-time emotion detection and response, which reduces user interaction and system effectiveness.The user interface in many existing systems is not user-friendly, making it difficult for users to interact with the system efficiently.

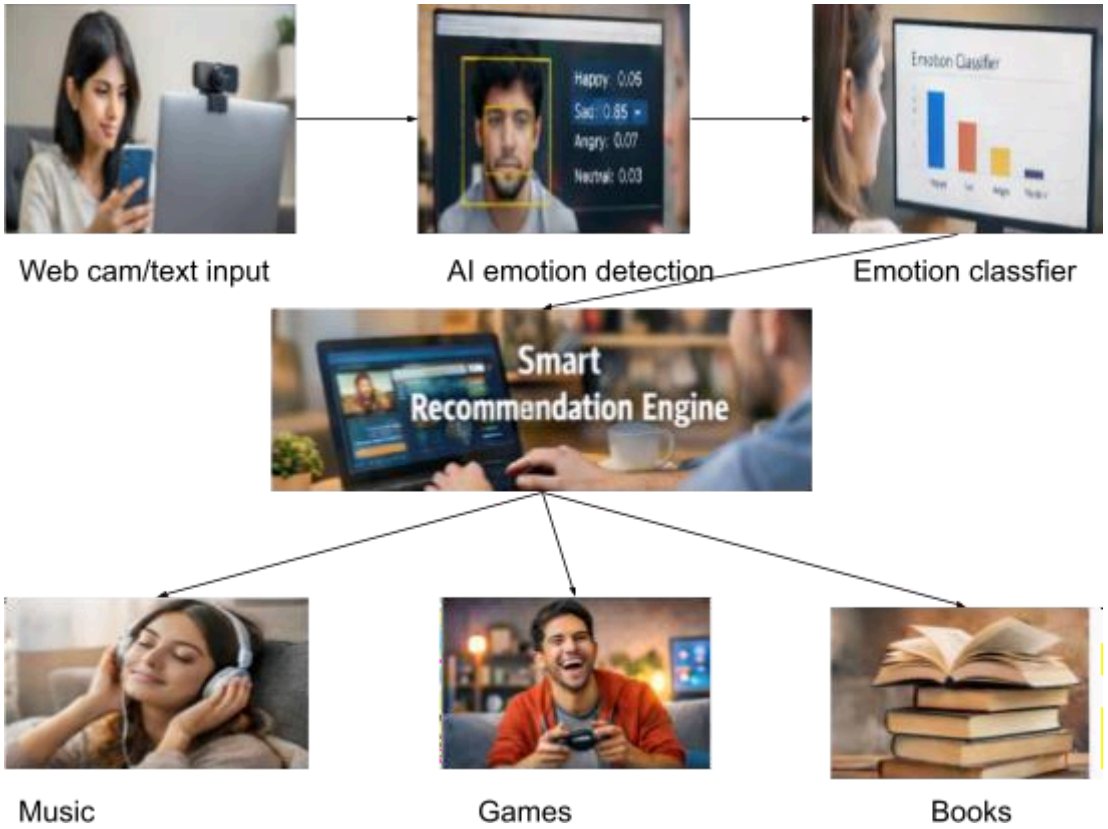

Figure 5.2 Proposed System Architecture

CHAPTER 6

SYSTEM REQUIREMENTS

6.1 HARDWARE REQUIREMENTS

- **Processor: Intel Core i3 higher**

  The processor is the main component that performs all system operations. A processor with good speed helps execute the application smoothly and handle multiple tasks efficiently. A higher-speed processor improves system responsiveness, reduces processing time, and supports multitasking capabilities, which is essential for handling secure data communication and machine learning operations in the proposed framework.

- **RAM: Minimum 4 GB**

  RAM is used to store temporary data while the system is running. A minimum of 4 GB RAM is required to ensure smooth performance of the messaging application. Adequate RAM helps in faster data processing, efficient handling of multiple tasks, and better performance of background operations such as security monitoring and data processing.

- **Hard Disk: 500 GB or above**

  A The hard disk is used to store system files, application software, and user data. Sufficient storage capacity helps maintain database records and system data. Sufficient storage space ensures that the system can maintain historical records, support data analysis, and provide reliable data retrieval without storage constraints.

- **Monitor: 15” or higher**

  The monitor is used to display the graphical interface of the system. A monitor with good resolution helps users interact with the application easily. A good display enhances user experience by making it easier to read information, navigate the system, and monitor application activities effectively.

- **Keyboard: Standard Keyboards**

  The keyboard is used to enter text and commands into the system. It allows users to type messages, login details, and other input information. A standard keyboard ensures ease of use, accuracy, and efficiency in data entry operations within the system.

- **Mouse: Optical Mouse**

  The mouse helps users navigate through the system interface. It allows easy selection of options, buttons, and menus in the application. The use of an optical mouse improves user control and enhances the overall usability of the application by providing precise and smooth navigation.

- **Web Camera**

  A HD webcam is required to capture the user’s facial expressions in real time for emotion detection using the CNN model.

6.2 SOFTWARE REQUIREMENTS

- **Operating System**

  The operating system provides the platform to run the application. Windows 10 or higher is required to support the development environment and ensure smooth execution of the system. It supports

various programming environments, libraries, and tools required for web development and machine learning integration.

The operating system also manages system resources such as memory, processor usage, and file handling, ensuring smooth functioning of the application. Additionally, it provides built-in security features like firewall and user authentication, which contribute to the overall protection of the system.

- **Image Processing Library**

  OpenCV is used for capturing images from the webcam and detect face.

- **Machine Learning Libraries**

  Libraries such as TensorFlow, Keras, and PyTorch are used to implement the Convolutional Neural Network (CNN) model for emotion detection. In this framework, these libraries can be used for tasks such as anomaly detection, pattern recognition, and identifying suspicious activities in messaging behavior.

  They also support efficient handling of large datasets and provide pre-built functions that simplify the implementation of complex algorithms, thereby improving system performance and accuracy.

- **Database: MySQL / SQLite**

  The database is used to store user details, encrypted messages, and system data. It helps manage and organize the data required for the messaging application. The database ensures data consistency, integrity, and quick retrieval of information. It also supports security features such as user authentication and access control to prevent unauthorized data access.

- **Frontend Technologies: HTML, CSS, JavaScript**

  Frontend technologies are responsible for creating the visual and interactive components of the application. HTML is used to design the structure of web pages, CSS is used for styling and layout design, and JavaScript is used to add dynamic behavior and interactivity. These technologies enable the development of responsive and user-friendly interfaces for functionalities such as user registration, login, and messaging. A well-designed frontend improves user experience, ensures easy navigation, and enhances the overall usability of the system.

- **Development Tool: Visual Studio Code**

  Visual Studio Code is a powerful and lightweight code editor used for developing the application. It provides various features such as syntax highlighting, code completion, debugging tools, and support for multiple programming languages. It also offers a wide range of extensions that help in improving productivity, such as Git integration, Python support, and live server functionality. This tool enables developers to write, test, and debug code efficiently, manage project files effectively, and streamline the overall development process.

CHAPTER 7

SYSTEM IMPLEMENTATIONS

7.1 LIST OF MODULES

- User Input Module
- AI Emotion Detection Module
- Emotion Classification Module
- Smart Recommendation Engine Module
- Activity Recommendation Module
- User Data and analytics Module

7.2 MODULE DESCRIPTION

7.2.1 User Input Module

The User Input Module is the first stage of the system where the user provides input to the application. The system collects input through two main methods: face camera and chatbot text input. In the face camera method, the webcam captures the user's facial image in real time. This image is used to analyze the user's facial expressions for emotion detection. In the chatbot input method, the user can type their feelings or mood into the system. The input provided by the user is processed and forwarded to the next stage of the system. This module ensures that the system receives accurate user data for further analysis. It acts as the interface between the user and the emotion detection system. The module is designed to be simple and user-friendly. Proper input collection helps improve the accuracy of emotion detection and recommendations.

7.2.2 AI Emotion Detection Module

The AI Emotion Detection Module is responsible for detecting emotions from the user's facial expressions. This module uses a Convolutional Neural Network (CNN) model to analyze facial images captured through the camera. The CNN model processes the image and extracts important facial features such as eyes, mouth, and facial structure. These features help identify emotional expressions. The model is trained using a dataset containing various facial expressions. The training process enables the system to recognize emotions accurately. This module plays an important role in converting raw image data into meaningful emotional information. It improves the system's ability to understand the user's emotional state. The detected emotion is then passed to the emotion classification module for further processing.

Furthermore, this module performs several preprocessing operations before the image is analyzed by the CNN model. The captured facial image is first converted into a suitable format for processing. Techniques such as image normalization, resizing, and feature enhancement are applied to improve detection accuracy. The CNN model then extracts important facial features by analyzing multiple layers of the neural network. Each layer identifies different patterns such as edges, facial structure, and expression details. These extracted features help the model understand subtle emotional differences between facial expressions. The system also uses trained datasets to improve the learning capability of the CNN model. As more data is processed, the model becomes more accurate in detecting emotions. This module plays a critical role in ensuring reliable emotion detection. Accurate detection is essential for generating meaningful activity recommendations in the next stages of the system.

7.2.3 Emotion classification Module

The Emotion Classification Module classifies the detected facial expressions into specific emotional categories. The emotions may include happiness, sadness, anger, surprise, and neutral mood. This module analyzes the features extracted by the CNN model and assigns the appropriate emotion label. The classification process ensures that the system correctly understands the user's emotional state. Accurate classification is essential for generating suitable activity recommendations. The module helps the system differentiate between multiple emotional expressions. It also ensures that the emotional data is organized and structured for further analysis. The classified emotion is then forwarded to the smart recommendation engine. This module improves the overall accuracy and efficiency of the system.

Furthermore, this module evaluates the probability scores generated by the emotion detection model to determine the most accurate emotional category. The classifier compares different emotional features and selects the emotion with the highest confidence level. This process helps improve the reliability of the classification results. The system may also use predefined thresholds to ensure that uncertain predictions are handled carefully. In some cases, if the emotion is not clearly identified, the system may classify it as a neutral state. The classified emotional data is then structured and prepared for the recommendation process. This module ensures that the emotional output is consistent and meaningful for further analysis. Accurate classification helps the system generate more appropriate and personalized activity suggestions. It also improves the overall efficiency of the emotion-based recommendation system.

7.2.4 Smart Recommendation Engine module

The Smart Recommendation Engine is responsible for generating personalized activity suggestions based on the user's emotional state. After

receiving the classified emotion from the previous module, the system analyzes it to determine suitable activities. The recommendation engine uses predefined rules and intelligent algorithms to generate appropriate suggestions. The goal is to improve the user's mood and emotional well-being. This module acts as the decision-making component of the system. It processes emotional information and converts it into meaningful recommendations. The system ensures that the recommendations match the user's current mood. The smart recommendation engine also helps improve user engagement. It plays a central role in connecting emotion detection with activity suggestions.

Furthermore, the smart recommendation engine uses the detected emotional information to determine the most suitable activities for the user. The engine may apply rule-based logic or simple recommendation algorithms to match emotions with appropriate activities. For example, if the user is detected as sad, the system may recommend relaxing music or motivational books. If the user is happy, the system may suggest entertaining games or energetic music. The recommendation engine also considers previously stored user preferences when generating suggestions. This helps in improving the personalization of recommendations. The system may update recommendations dynamically as the user's emotional state changes.

7.2.5 Activity Recommendation Module

The Activity Recommendation Module provides different types of activities based on the user's emotional state. These activities include music therapy, mood games, yoga and meditation, and book reading suggestions. Music therapy helps users relax and improve their mood through suitable songs. Mood games provide entertainment and help reduce stress. Yoga and meditation suggestions help users calm their mind and improve mental health. Book recommendations provide motivational or entertaining reading options.

The activities are selected based on the emotion detected by the system. This module aims to enhance the user's emotional balance and well-being. The recommendations help users engage in positive activities that match their emotional condition.

Furthermore, this module organizes the recommended activities into different categories so that users can easily choose from them. Each category is designed to help the user improve or maintain their emotional well-being. For instance, music therapy can help reduce stress and improve mood. Mood games provide entertainment and help users relax their minds. Yoga and meditation suggestions support mental calmness and emotional balance. Book recommendations provide educational, motivational, or entertaining reading options. The module ensures that the suggested activities are relevant to the detected emotion. It also allows users to select activities based on their interests. By offering multiple activity options, the system enhances user engagement and satisfaction. This module plays an important role in providing positive and helpful experiences for users.

7.2.6 User Data and Analytics Module

The User Data and Analytics Module stores and analyzes the user's emotional data and activity history. The system records the detected emotions and the activities suggested to the user. This data is used to understand user behavior and emotional trends over time. The module compares current emotional states with previous data to identify patterns. Graphical visualization such as charts and graphs can be used to represent this information. The analytics help evaluate how the recommendations influence the user's mood. This module helps improve the system by providing insights into user preferences. It also supports future improvements in the recommendation process. The analytics feature makes the system more intelligent and adaptive.

Furthermore, this module manages the interaction between the user and the system after the recommendations are displayed. It allows users to view, select, and engage with the suggested activities based on their preferences. The system provides an interactive interface where users can easily navigate through different recommendations such as music, games, yoga, meditation, or books. In addition, the module collects user feedback regarding the usefulness of the recommended activities. This feedback helps the system understand user satisfaction and improve future recommendations. The module may also track user selections and usage patterns to refine the recommendation process. By analyzing this feedback, the system can adapt its suggestions to better match the user's emotional needs. This continuous interaction ensures that the system remains user-friendly and responsive. Ultimately, the module enhances the overall effectiveness and personalization of the emotion-based activity recommendation system.

CHAPTER 8

SYSTEM TESTING

8.1 USER INTERFACE TESTING

User Interface (UI) testing focuses on evaluating the usability and visual design of the system interface. The main objective of this testing is to ensure that users can interact with the system easily and understand the system features without confusion.

In this project, UI testing checks whether the system interface clearly displays the webcam input, chatbot input options, and recommended activities. It also verifies that all buttons, icons, and menus function correctly. The layout should be simple, attractive, and easy for users to navigate

UI testing also ensures that the system provides clear feedback to the user. For example, when the system detects an emotion, the result should be displayed clearly along with the recommended activities. A well-designed user interface improves user satisfaction and enhances the overall effectiveness of the system.

8.2 INTEGRATION TESTING

Integration testing is conducted to verify that different modules of the system interact with each other correctly. In large systems, modules are developed separately, and integration testing ensures that when these modules are combined, they work together without errors.

In the Emotion-Based Activity Recommendation System, integration testing checks the communication between modules such as the input capture module, CNN processing module, emotion classification module, smart recommendation engine, and activity recommendation module. This testing ensures that the output of one module becomes the correct input for the next

module.

For example, the emotion detection module sends facial feature data to the CNN model, which processes the information and forwards it to the emotion classifier. The classifier then passes the identified emotion to the recommendation engine, which generates suitable activity suggestions. Integration testing ensures that this entire data flow occurs smoothly and without interruptions.

8.3 SYSTEM TESTING

System testing evaluates the complete and fully integrated system to ensure that it meets the specified requirements. In this stage, the entire messaging platform is tested to verify its functionality, reliability, and performance. All modules such as user authentication, message request handling, encryption, passkey generation, machine learning monitoring, and database management are tested together. The testing ensures that the system behaves correctly in real-time scenarios. For example, when a sender sends a message, the receiver should receive a request notification, accept the request, and then view the encrypted message.

8.4 PERFORMANCE TESTING

Performance testing evaluates the speed, efficiency, and responsiveness of the system. The objective of this testing is to ensure that the system performs efficiently even when processing real-time data.

In the Emotion-Based Activity Recommendation System, performance testing checks how quickly the system detects facial expressions and generates recommendations. Since emotion detection is a real-time process, the system must respond quickly without noticeable delays.

This testing also measures the system’s ability to process continuous

webcam input and maintain stable performance. If the system takes too long to detect emotions or generate recommendations, it may affect the user experience. Performance testing helps identify such issues and improves the system's overall efficiency.

8.5 ACCURACY TESTING

Accuracy testing is conducted to measure how accurately the system detects and classifies user emotions. Since the system uses a Convolutional Neural Network (CNN) for emotion detection, it is important to evaluate the model's prediction accuracy.

During accuracy testing, the system is tested using different facial expressions such as happy, sad, angry, surprised, and neutral. The results produced by the system are compared with the actual emotions to determine the accuracy rate.

If the system incorrectly classifies emotions, adjustments can be made to improve the training dataset or model parameters. High accuracy ensures that the recommendations provided by the system are relevant and useful for the user.

8.6 FUNCTIONAL TESTING

Functional testing is performed to verify that each function of the system operates according to the specified requirements. The main purpose of this testing is to ensure that all the features of the Emotion-Based Activity Recommendation System work correctly. During functional testing, the system is tested by providing different types of inputs and checking whether the expected output is produced.

In this project, functional testing verifies several operations such as capturing user input through the webcam or chatbot, detecting facial expressions, processing the input through the CNN model, and classifying the emotions accurately. It also ensures that once the emotion is detected, the

system generates appropriate recommendations like music, mood games, yoga or meditation activities, and book suggestions.

Functional testing also confirms that the user interface elements such as buttons, menus, and activity selection options operate properly. If any function does not perform as expected, the issue is identified and corrected. This testing ensures that the system behaves exactly as designed and fulfills the functional requirements of the project.

CHAPTER 9

RESULTS AND DISCUSSION

The Emotion-Based Activity Recommendation System was successfully developed and tested to analyze the emotional state of users and provide suitable activity recommendations. The system integrates multiple modules including input capture through webcam or chatbot, emotion detection using a Convolutional Neural Network (CNN), emotion classification, and a smart recommendation engine. The main objective of the system is to identify the user's emotional condition and suggest activities that help improve or maintain their emotional well-being.

During the testing phase, the system was able to detect various facial expressions such as happy, sad, angry, and neutral. The CNN model processed the facial features extracted from the input image and classified the emotional state with satisfactory accuracy. Once the emotion was identified, the smart recommendation engine generated appropriate activity suggestions such as listening to music, playing mood-based games, reading books, or performing yoga and meditation.

The results show that the system can effectively detect emotions in real time and provide meaningful recommendations based on the detected emotional state. The integration between different modules worked smoothly, ensuring a continuous flow of data from emotion detection to recommendation generation. The user interface also allowed users to easily interact with the system and view recommended activities without difficulty.

Overall, the proposed system demonstrates the practical application of emotion recognition and recommendation technologies in improving user well-being. Although the system performs effectively, future improvements may include expanding the emotion dataset, enhancing model accuracy, and

incorporating more personalized recommendation techniques. These improvements can further increase the reliability and usefulness of the system in real-world applications.

CHAPTER 10

CONCLUSION AND FUTURE WORK

10.1 CONCLUSION

The Emotion-Based Activity Recommendation System was successfully designed and implemented to identify the emotional state of users and provide suitable activity recommendations. The system combines emotion detection techniques with intelligent recommendation methods to support users in managing their emotional well-being. By using webcam or chatbot input, the system captures user data and processes it through a Convolutional Neural Network (CNN) model to detect facial emotions accurately.

The detected emotions are then classified and passed to the smart recommendation engine, which suggests activities such as music, mood games, yoga, meditation, and book reading. These activities are selected based on the user's emotional condition to help improve mood, reduce stress, and promote mental relaxation. The integration of different modules ensures smooth communication between the emotion detection and recommendation processes.

The results obtained from the system demonstrate that emotion recognition technology can be effectively used to provide personalized activity suggestions. The system performs efficiently in real-time emotion detection and delivers meaningful recommendations that can positively influence the user's emotional state.

10.2 FUTURE WORK

Although the Emotion-Based Activity Recommendation System performs effectively in detecting emotions and suggesting suitable activities, several improvements can be made in the future to enhance the system's performance and usability.

In the future, the system can be improved by using larger and more diverse datasets to train the emotion detection model. This will help increase the accuracy of emotion recognition and enable the system to detect a wider range of emotional expressions. Advanced deep learning techniques and improved CNN architectures can also be implemented to enhance the reliability of emotion classification.

Another possible improvement is the integration of additional input methods such as voice recognition and text sentiment analysis. By analyzing the user's speech tone or written messages, the system can better understand the user's emotional state and provide more accurate recommendations.

In addition, the system can be developed as a mobile application or web-based platform so that users can access the system easily from different devices. More activity categories such as stress management exercises, mental health tips, and motivational content can also be included to improve the usefulness of the system.

Overall, these future improvements will help make the Emotion-Based Activity Recommendation System more intelligent, accurate, and beneficial for supporting users' emotional well-being.

APPENDIX - A

SOURCE CODE

```
<!DOCTYPE html>

<html>

<head>

<title>Emotion Detection System</title>

<style> body{ margin:0;

font-family:Arial;

background:linear-gradient(135deg,#667eea,#764ba2,#ff758c);
height:100vh;

}

/* CHATBOT BUTTON */

.chatbot-btn{ position:fixed; top:20px; right:20px; background:#ff4b5c;
color:white; border:none; padding:15px; border-radius:50%;
font-size:22px; cursor:pointer;

}

/* CHATBOX */

.chatbox{ position:fixed; top:80px;
```

```
border-radius:10px;

box-shadow:0px 10px 25px rgba(0,0,0,0.3); display:none;

flex-direction:column;

}

/* HEADER */

.chat-header{ background:#764ba2; color:white; padding:10px;

border-radius:10px 10px 0 0; display:flex;

align-items:center;}

.chat-header img{ width:30px; margin-right:10px;

}

/* CHAT MESSAGES */

.chat-messages{ height:250px; overflow-y:auto; padding:10px;
font-size:14px;

(request.form['name'], request.form['password'], role)).fetchone()

if user:

session['user'], session['org'], session['role'] = user['name'], user['org'],
user['role']

return jsonify({"status": "success"})

else:
```

```
}

/* INPUT AREA */

.chat-input{ display:flex;

border-top:1px solid #ddd;

}

.chat-input input{ flex:1; padding:10px; border:none; outline:none;

}

.chat-input button{ background:#ff4b5c; border:none; color:white;
padding:10px; cursor:pointer;

}

/* MAIN PAGE */

.container{ display:flex

flex-direction:column; justify-content:center; align-items:center; height:100vh;
color:white;

text-align:center;

}

.start-btn{ background:#ffcc00; color:black; padding:15px 40px;
font-size:20px; border:none;

border-radius:10px; cursor:pointer; margin-top:20px;
```

```
}
</style>
</head>
<body>
<!-- CHATBOT BUTTON -->
<button class="chatbot-btn" onclick="toggleChat()">🤖</button>
<!-- CHATBOT -->
<div class="chatbox" id="chatbox">
<div class="chat-header">
<img src="https://cdn-icons-png.flaticon.com/512/4712/4712027.png"> AI Emotion Assistant
</div>
<div class="chat-messages" id="messages">
<p>🤖 Hello! Tell me how you feel.</p>
</div>
<div class="chat-input">
<input type="text" id="userInput" placeholder="Type or speak your emotion...">
<button onclick="startVoice()">🎤</button>
```

```
<button onclick="sendMessage()">Send</button>

</div>

</div>

<!-- MAIN PAGE -->

<div class="container">

<h1>AI Emotion Detection System</h1>

<p>Detect emotions and get personalized suggestions</p>

<a href="/detect">

<button class="start-btn">Start Detection</button>

</a>

</div>

<script>

function toggleChat(){

var chat=document.getElementById("chatbox");
if(chat.style.display=="none" || chat.style.display==""){
chat.style.display="flex";

}
```

```
recognition.lang = "en-US"; recognition.start(); recognition.onstart =
function(){

console.log("Voice recognition started");

}

recognition.onresult = function(event){ var speech =
event.results[0][0].transcript;

document.getElementById("userInput").value = speech; sendMessage();

}

}

function speak(text){

const speech = new SpeechSynthesisUtterance(); speech.text = text;

speech.lang = "en-US"; window.speechSynthesis.speak(speech);

},

</script>

</body>

</html>
```

APPENDIX – B

SCREENSHOTS

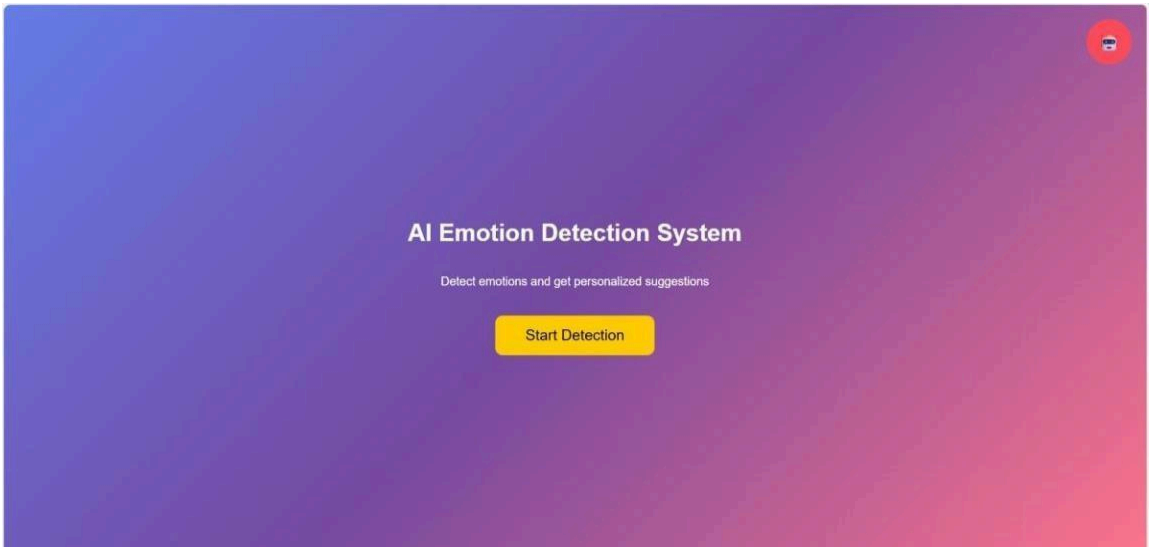

Figure B.1 Start Detection

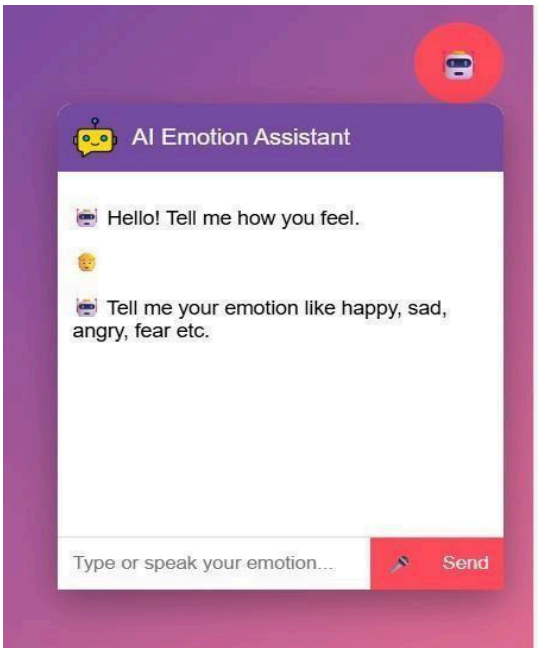

Figure B.2 Text Input

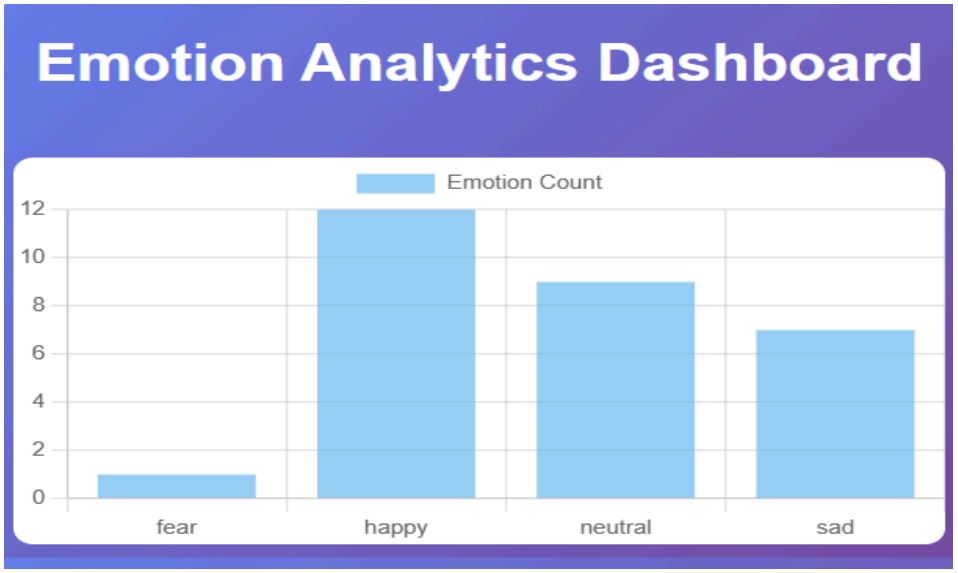

Figure B.3 Emotion Analysis

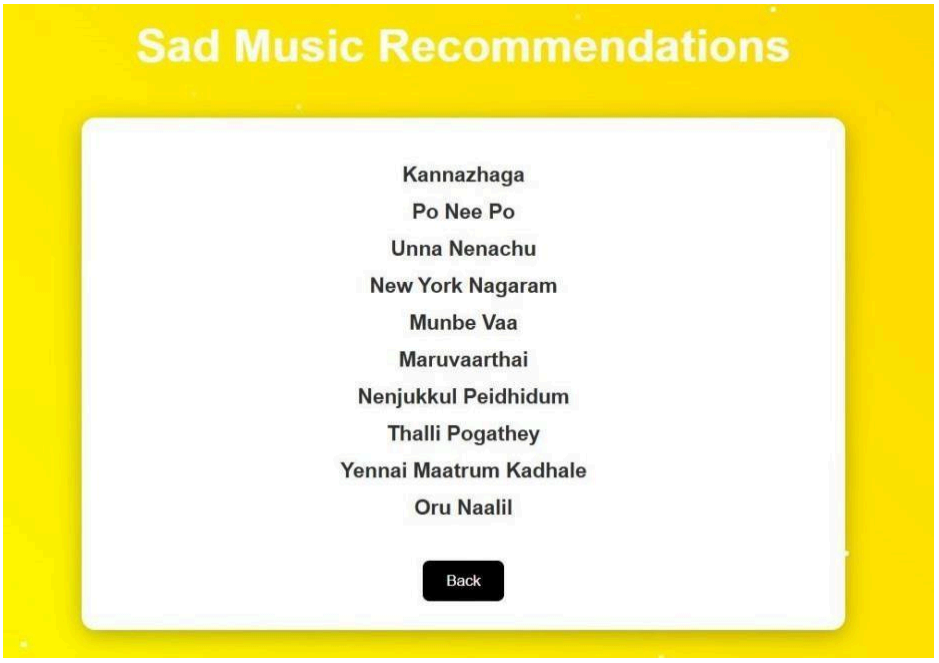

Figure B.4 Music Recommendation

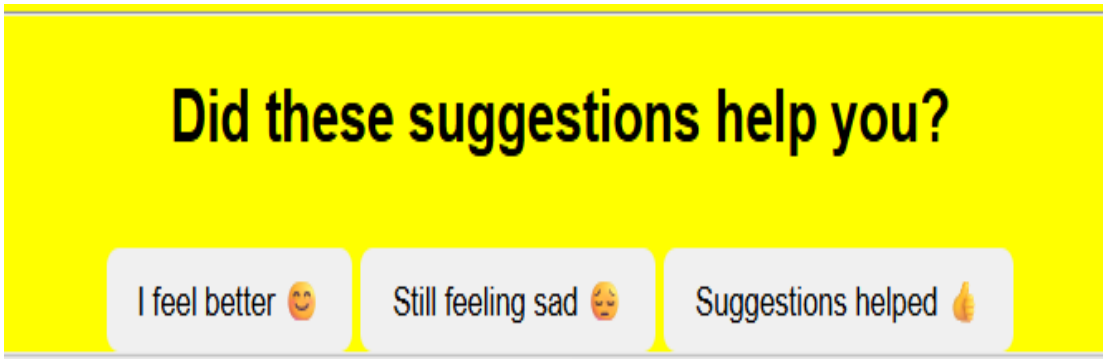

Figure B.5 Feedback

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10. Smart Emotion-Based Activity Recommendation System – S. W Y. Chen, and L. Zhang.

CERTIFICATES

Stay Ahead

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Department of CSE, IT, CT & ISE

Certificate of Participation

This is to certify that Mr./Ms. DHARSHINI S S from K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY has presented a paper titled AI- BASED REAL TIME EMOTION RECOGNITION AND PERSONALIZED MUSIC AND ACTIVITY RECOMMENDATION SYSTEM in SIXTH INTERNATIONAL CONFERENCE ON ARTIFICIAL INTELLIGENCE AND SMART COMPUTING (ICAISC'26) held during 19th - 21st March 2026 at Bannari Amman Institute of Technology, Sathyamangalam.

Organizing Secretary | Convenor | Principal

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